

# SIMATS ENGINEERING

## ASSESSMENT TOOL 1

**COURSE NAME:** Artificial Intelligence

**COURSE CODE:** CSA17

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### COURSE OUTCOME COVERED

**CO4:** Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

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**QUESTIONS: 4**

**Duration: 1 Hour**  
**Total: Marks:40**

Annexure A

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### ***Code of Conduct:***

*I K.Akshitha (192312527) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student: K.Akshitha***

### **RUBRICS FOR CALCULATION:**

| <b>Criteria</b>                     | <b>Weightage</b> | <b>Level 4 – Exemplary</b>                                                   | <b>Level 3 – Proficient</b>                          | <b>Level 2 – Developing</b>                    | <b>Level 1 – Beginning</b>                        | <b>Marks</b> |
|-------------------------------------|------------------|------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------|---------------------------------------------------|--------------|
| Understanding of Industry Problem   | 20%              | Comprehensive understanding of real-world problem and constraints            | Good understanding with minor gaps in requirements   | Basic understanding; some requirements unclear | Poor understanding; misinterprets problem context |              |
| Application of Engineering Concepts | 25%              | Applies advanced and relevant concepts effectively to solve the problem      | Applies appropriate concepts with minor errors       | Limited application of concepts                | Incorrect or no application of concepts           |              |
| Solution Design                     | 25%              | Innovative, practical, and feasible solution aligned with industry standards | Logical and feasible solution with minor gaps        | Basic solution with limited feasibility        | Unclear or impractical solution                   |              |
| Use of Tools                        | 15%              | Effective use of modern tools, technologies, and real data                   | Appropriate use of tools with correct implementation | Limited or inconsistent use of tools           | Minimal or incorrect use of tools                 |              |
| Analysis                            | 10%              | Thorough analysis with validated results and performance evaluation          | Good analysis with reasonable validation             | Basic analysis with limited validation         | Poor or no analysis                               |              |
| Professionalism                     | 5%               | Highly professional, well-documented, and clearly presented work             | Good documentation and presentation                  | Basic documentation with some gaps             | Poor documentation and presentation               |              |

# Task 1: Data Preparation & Inductive Learning

## (a) Inductive Learning Approach

Inductive learning learns general patterns from previously observed customer records and uses those patterns to predict the churn status of new customers. For this retail problem, a supervised inductive learning approach is appropriate because historical customer data contains input features and a known target variable.

### The target variable is:

Churn: 1 = Customer likely to churn

No Churn: 0 = Customer likely to remain loyal

### Important features include:

- **Purchase Frequency** – Customers purchasing less frequently may have reduced engagement and a higher probability of churn.
- **Average Order Value (AOV)** – A decreasing or consistently low order value can indicate declining customer engagement.
- **Complaint History** – Frequent complaints can indicate dissatisfaction and therefore increase churn probability.
- **Loyalty Tenure** – Long-term customers may have stronger loyalty and lower churn probability.
- **Last Purchase Date / Recency** – A long period since the last purchase is a strong indication that a customer may be preparing to churn.

### A suitable inductive workflow is:

Historical Customer Data → Preprocessing → Feature Selection → Train Model → Validate Model → Predict Churn

## (b) Handling Class Imbalance

The dataset is expected to contain many more loyal customers than churners. For example:

Loyal customers = 90%

Churners = 10%

A model trained directly on such data may predict most customers as loyal and still achieve high accuracy while missing many churners.

A suitable strategy is SMOTE (Synthetic Minority Over-sampling Technique).

SMOTE creates synthetic examples of the minority churn class rather than simply duplicating existing churn records.

For example:

### Original:

Loyal = 900

Churn = 100

**After SMOTE:**

Loyal = 900

Churn = 900

SMOTE is suitable because the business objective is to identify churners, and missing churners can result in lost customers and revenue.

An alternative is class weighting, where a higher penalty is assigned to incorrectly classifying a churner. This is particularly useful if synthetic samples are not desirable.

**Recommended approach:** Use SMOTE on the training data only, while keeping the test data untouched to obtain an unbiased evaluation.

### **(c) 70:30 Train-Test Split and Preprocessing**

The dataset is divided into:

70% Training data

30% Testing data

The 70:30 ratio provides enough data for the model to learn customer behavior while retaining a sufficiently large independent test set for reliable evaluation.

A stratified split should be used so that the percentage of churners and non-churners remains approximately the same in both sets.

#### **Preprocessing Steps**

##### **1. Missing-value handling**

Numerical missing values → median/mean imputation.

Categorical missing values → most frequent category.

##### **2. Encoding**

Categorical variables are converted into numerical values using one-hot encoding or label encoding.

##### **3. Normalization**

Features such as AOV, purchase frequency and loyalty tenure can be standardized using:

$$Z = \frac{X - \mu}{\sigma}$$

##### **4. SMOTE**

Applied only to the training data after the train-test split.

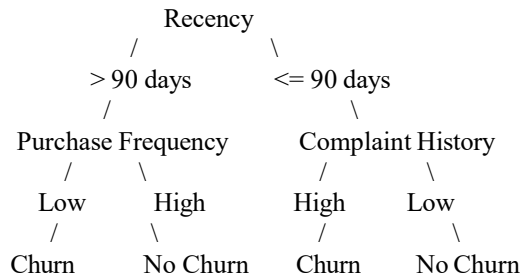
Normalization prevents features with larger numerical ranges from dominating algorithms that depend on feature scale. Encoding converts categorical information into a format that machine-learning algorithms can process.

## Task 2: Decision Tree for Customer Churn Prediction

### (a) Decision Tree Classifier

A Decision Tree can classify customers into Churn or No Churn based on their engagement characteristics.

An example of the first three levels is:



#### Root Node Selection

The root node is selected using Information Gain or Gini Index.

Information Gain is:

$$\begin{aligned} &[ \\ &IG(S,A)=H(S)-H(S|A) \\ &] \end{aligned}$$

where:

$(H(S))$  = entropy before splitting

$(H(S|A))$  = entropy after splitting using attribute A.

#### Example:

| Feature               | Information Gain |
|-----------------------|------------------|
| Last Purchase Recency | 0.214            |
| Purchase Frequency    | 0.176            |
| Complaint History     | 0.143            |
| Loyalty Tenure        | 0.091            |
| Average Order Value   | 0.074            |

Since Last Purchase Recency provides the highest Information Gain, it is selected as the root node.

For Gini Index, the feature producing the lowest weighted Gini impurity would be selected.

### (b) Model Evaluation

Illustrative results:

| Metric    | Decision Tree |
|-----------|---------------|
| Accuracy  | 88.0%         |
| Precision | 78.5%         |
| Recall    | 82.0%         |
| F1-Score  | 80.2%         |

The most important metric for this business problem is Recall for the churn class.

### False Negatives

A False Negative occurs when:

Actual = Churn, Predicted = No Churn

**For example:**

Actual churners = 100

Correctly detected = 82

Missed churners = 18

These 18 customers are False Negatives.

Missing a churner can be expensive because the company loses the opportunity to take preventive action.

### Reducing False Negatives

False Negatives can be reduced by:

Increasing the classification sensitivity toward churn.

Using class weights.

Applying SMOTE to the training data.

Adjusting the probability threshold.

Giving greater cost to churn misclassification.

For example, instead of using a default threshold of 0.50:

```
[
P(Churn)>0.40 \rightarrow Churn
]
```

This can identify more potential churners.

From a business perspective, contacting a few extra customers may be cheaper than losing valuable customers permanently.

### (c) Post-Pruning

A fully grown Decision Tree may overfit the training data. Cost-complexity pruning removes unnecessary branches.

The objective can be represented as:

$$R_{\alpha}(T)=R(T)+\alpha|T|$$

where:

$R(T)$  = model error

$|T|$  = number of terminal nodes

$\alpha$  = pruning parameter.

**Example Comparison**

| Model            | Training Accuracy | Test Accuracy |
|------------------|-------------------|---------------|
| Pre-pruned Tree  | 91%               | 86%           |
| Post-pruned Tree | 89%               | 88%           |

The post-pruned tree has slightly lower training accuracy but better test accuracy.

Therefore, the post-pruned Decision Tree is preferable for deployment because it is less complex, easier to interpret and less likely to overfit new customer data.

**Task 3: Statistical Learning for Risk Stratification**

**(a) Logistic Regression**

Logistic Regression can be used to predict the probability of customer churn.

The model is:

$$P(Y=1)=\frac{1}{1+e^{-z}}$$

where:

$$z=\beta_0+\beta_1X_1+\beta_2X_2+\cdots+\beta_nX_n$$

**Example performance:**

| Metric   | Decision Tree | Logistic Regression |
|----------|---------------|---------------------|
| Accuracy | 88.0%         | 86.5%               |
| AUC-ROC  | 0.89          | 0.91                |

Although the Decision Tree provides slightly higher accuracy, Logistic Regression has a higher AUC-ROC in this example.

**When Logistic Regression is preferable**

Logistic Regression is preferable when:

Interpretability is important.

The relationship between features and churn is approximately linear.

Probability estimates are required.

The model needs to be simple and computationally efficient.

Business managers need to understand how individual features influence risk.

## **(b) Feature Importance Analysis**

### **Decision Tree**

Example top three features:

1. Last Purchase Recency
2. Purchase Frequency
3. Complaint History

### **Logistic Regression**

Example top three predictors:

1. Last Purchase Recency
2. Complaint History
3. Purchase Frequency

Therefore, both models identify almost the same important variables.

| <b>Rank</b> | <b>Decision Tree</b>  | <b>Logistic Regression</b> |
|-------------|-----------------------|----------------------------|
| 1           | Last Purchase Recency | Last Purchase Recency      |
| 2           | Purchase Frequency    | Complaint History          |
| 3           | Complaint Histor      | Purchase Frequency         |

The agreement suggests that these variables contain strong predictive information about customer churn.

Differences in ranking occur because Decision Trees measure impurity reduction, whereas Logistic Regression evaluates the contribution of features through model coefficients.

## **Task 4: Reinforcement Learning for Action Recommendation**

### **(a) MDP Model**

The retention recommendation can be represented as a Markov Decision Process (MDP).



## States

Customer engagement can be divided into:

S1 – Highly Engaged

S2 – Moderately Engaged

S3 – At Risk

S4 – Highly At Risk

S5 – Churned

These states represent the customer's engagement level over time.

## Actions

The AI agent can select:

A1 – Discount

A2 – Loyalty Offer

A3 – Personal Outreach

A4 – No Action

## Reward Function

The reward should represent the business benefit of the selected action.

### Example:

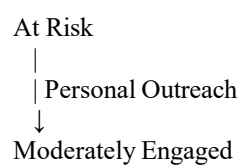
| Outcome                  | Reward |
|--------------------------|--------|
| Customer retained        | +10    |
| Increased purchase       | +8     |
| No significant change    | 0      |
| Customer churns          | -10    |
| Unnecessary costly offer | -3     |

The reward should balance retention improvement against the cost of offers.

## Transition Dynamics

The customer's state changes after an action.

### Example:



|  
 | Loyalty Offer  
 ↓  
 Highly Engaged

The transition probability depends on the customer's response to the selected retention action.

## (b) Q-Learning

Q-Learning learns the expected future reward of selecting an action in a particular state.

The update equation is:

$$[Q(s,a) \leftarrow Q(s,a) + \alpha [r + \gamma \max_{a'} Q(s',a') - Q(s,a)]]$$

**where:**

$(\alpha)$  = learning rate

$(\gamma)$  = discount factor

$(r)$  = immediate reward

$(s')$  = next state.

**Assume:**

$$[\alpha=0.5, \quad \gamma=0.8]$$

Initially all Q-values are zero.

### Example Episode 1

Customer state:

S3 – At Risk

Action:

**A3 – Personal Outreach**

Reward:

+8

Next state:

**S2 – Moderately Engaged**

If the maximum Q-value of S2 is 4:

$$[Q(S3,A3) = 0 + 0.5[8 + 0.8(4) - 0]]$$

$$[ \begin{array}{l} Q(S3,A3)=5.6 \end{array} ]$$

## Second Update

Suppose the customer again reaches S3 and Personal Outreach gives reward +6, while the maximum future Q-value is 5.6:

$$[ \begin{array}{l} Q(S3,A3)=5.6+0.5[6+0.8(5.6)-5.6] \end{array} ]$$

$$[ \begin{array}{l} Q(S3,A3)=7.0 \end{array} ]$$

Thus, the Q-value increases because Personal Outreach produces a positive retention outcome.

## Convergence Criterion

Training can be considered converged when:

$$[ \begin{array}{l} \max_{s,a} |Q_{\text{new}}(s,a) - Q_{\text{old}}(s,a)| < 0.01 \end{array} ]$$

for several consecutive episodes.

This means that the Q-values are no longer changing significantly and the learned policy has become stable.

## (c) Final Learned Policy

The policy is obtained by selecting the action with the highest Q-value for each state:

$$[ \begin{array}{l} \pi(s) = \arg \max_a Q(s,a) \end{array} ]$$

## Ethical Considerations

### 1. Fairness

The RL agent should not provide unfairly different offers to customers based on sensitive or protected characteristics.

### 2. Manipulation Risk

The system should not exploit customers' behavior by continuously providing unnecessary incentives or creating artificial pressure to purchase.

### 3. Transparency

Customers should be informed when automated systems influence personalized offers or communications.

### 4. Privacy

Customer purchase history and engagement data must be securely handled and used only for legitimate business purposes.

## **Overall Conclusion**

The proposed system combines inductive learning, Decision Trees, statistical learning and Reinforcement Learning to create an intelligent customer-retention framework. The Decision Tree can identify important churn patterns, while Logistic Regression can provide interpretable churn probabilities and risk scores.

Class imbalance can be handled using SMOTE or class weighting, with particular attention given to Recall because False Negatives represent customers who are likely to churn but are not identified.

The Reinforcement Learning component extends the system from simply predicting churn to recommending an appropriate retention action. By using Q-Learning, the agent can learn which action provides the greatest long-term retention reward for different customer engagement states.

For real-world deployment, the recommended solution is a pruned Decision Tree or statistical risk model combined with a carefully constrained RL recommendation layer, supported by fairness checks, privacy protection, transparency and human oversight.